



# CMS CAPILLARIES PRODUCTION FOR LHC HIGH LUMINOSITY UPGRADE

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## CAPILLARIES IN CMS (Compact Muon Solenoid)

With the **High Luminosity** LHC delivering up to 7.5 times more luminosity than the current LHC, CMS will upgrade its detectors to cope with harsher radiation, higher data rates, and increased pileup. To operate safely under these conditions, the upgraded silicon detectors require a highly efficient cooling system: since the increased radiation levels induce higher leakage currents in the sensors, the modules must be kept at **-35°C** to prevent thermal runaway. **Bi-phase CO<sub>2</sub>** cooling is employed to meet these strict thermal requirements, operating on a **2PACL** (2-Phase Accumulator Controlled Loop) cycle.

Within this 2PACL system, capillaries ensure :

- The **correct flow distribution** in each of the parallel branches of sub-detectors: each capillary guarantees about 80% of the total pressure drop of a detector loop, assuring that the neighboring loops can still function in case of circuit pressure unbalance.
- A well-defined pressure drop at the detector inlet, which is crucial for efficient cooling. The aim is to **trigger boiling** inside the evaporator of the detector, resulting in an improved heat transfer.

| Parameter             | Typical value  |
|-----------------------|----------------|
| Outer diameter (OD)   | 1.2 mm         |
| Inner diameter (ID)   | 0.5 - 0.8 mm   |
| Typical length        | 400- 2000 mm   |
| Material              | AISI 316L      |
| Operating pressure    | Approx. 35 bar |
| Temperature range     | 20°C to -40°C  |
| Number of capillaries | 2094           |

## ENGINEERING CHALLENGES

- SMALL DIMENSIONS**  
Precision handling of 1.2 mm capillaries
- LEAK TEST**  
Ensuring vacuum and pressure tightness
- CAPILLARIES PURITY**  
Preventing internal contamination
- TEMPERATURE**  
Maintaining -35°C temp. during testing
- BRAZING**  
Ensuring process repeatability while reducing failures

## PRODUCTION PROCESS

### RAW MATERIAL

Stainless steel pipes are supplied in large spools with an outer diameter of 1.2 mm and inner diameters ranging from 0.5 to 0.8 mm. Each spool is subjected to a metrological inspection to verify ID and OD tolerances.

### PROTOTYPING

Capillaries are manually cut from the spool, and both ends are polished to fully open the inner channel.

They are then formed into the final detector geometry to validate the required shape and length.

### EDM CUT

After defining the optimal geometry and length, the required number of capillaries is manually prepared with an additional +100 mm margin.

The capillaries are grouped into batches and transferred for precision EDM cutting to achieve smooth and repeatable end surfaces.

### TESTING

EDM-cut capillaries are installed in the COS (Capillary Open System) testing system and operated under controlled mass flow and temperature conditions. During the test, pressure drop (DP) stability is monitored. A stable operating period of 5 minutes is required for data validation. Each capillary has its own metal tag with a unique identification number.

### BRAZING

Validated capillaries are joined to the 1/8" VCR connectors via furnace or induction vacuum brazing (BFM: Microbraz 4778; maximum temperature cycle: 1060°C).

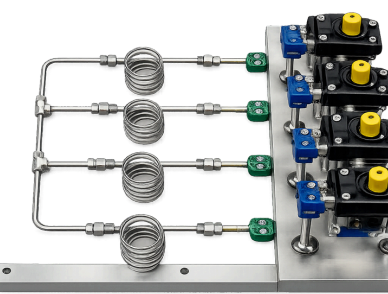
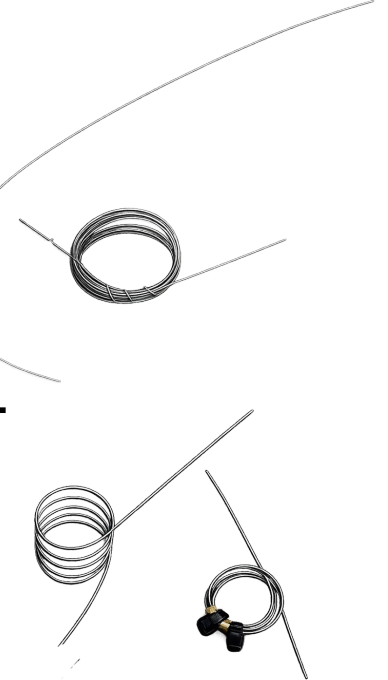
Since the connectors are suspended during brazing, the capillary ends are gently pre-bent to avoid any relative movement, ensuring a 2mm protrusion beyond the gland surface.

### JOINTS QUALITY CONTROL

Each assembled capillary undergoes pressure and leak testing to verify mechanical integrity and tightness. When required, X-ray inspection is additionally performed to validate the internal quality of the brazed joints and detect possible defects. Only components fulfilling acceptance criteria are approved for detector integration.

### DATA MANAGEMENT

All production parameters, testing results, brazing records and quality control measurements are stored in the CERN EDMS database. This ensures full traceability and long-term production monitoring for each capillary batch.



## COILING IMPACT ON DP

| CAP NAME        | GEOMETRY | ID (mm) | L (mm) | NUMBER OF LOOPS | FLOW (gr/s) | DP (bar) | DP/flow | DP difference | DP increase per loop (bar) |
|-----------------|----------|---------|--------|-----------------|-------------|----------|---------|---------------|----------------------------|
| HGCAL A         | STRAIGHT | 0,80    | 1250   | 14 loops        | 3,50        | 9,64     | 2,75    | 1,69 ↑        | 0,12                       |
|                 | COILED   |         |        |                 | 3,51        | 11,33    | 3,23    |               |                            |
| TBPX MT Z-L4-10 | STRAIGHT | 0,80    | 1000   | 10 loops        | 3,53        | 7,46     | 2,10    | 1,45 ↑        | 0,15                       |
|                 | COILED   |         |        |                 | 3,54        | 8,91     | 2,52    |               |                            |
| TEDD 1          | STRAIGHT | 0,60    | 890    | 8 loops         | 1,86        | 8,75     | 4,71    | 1,35 ↑        | 0,16                       |
|                 | COILED   |         |        |                 | 1,86        | 10,10    | 5,42    |               |                            |
| TEDD 2          | STRAIGHT | 0,60    | 1275   | 14 loops        | 1,59        | 8,84     | 5,56    | 1,72 ↑        | 0,12                       |
|                 | COILED   |         |        |                 | 1,59        | 10,56    | 6,63    |               |                            |
| TBPX MT Z-L2-8  | STRAIGHT | 0,50    | 860    | 6 loops         | 1,29        | 9,38     | 7,26    | 1,37 ↑        | 0,20                       |
|                 | COILED   |         |        |                 | 1,29        | 10,74    | 8,32    |               |                            |
| TBPX MT Z-L1-4  | STRAIGHT | 0,50    | 840    | 6 loops         | 1,28        | 9,31     | 7,26    | 1,21 ↑        | 0,23                       |
|                 | COILED   |         |        |                 | 1,28        | 10,52    | 8,20    |               |                            |

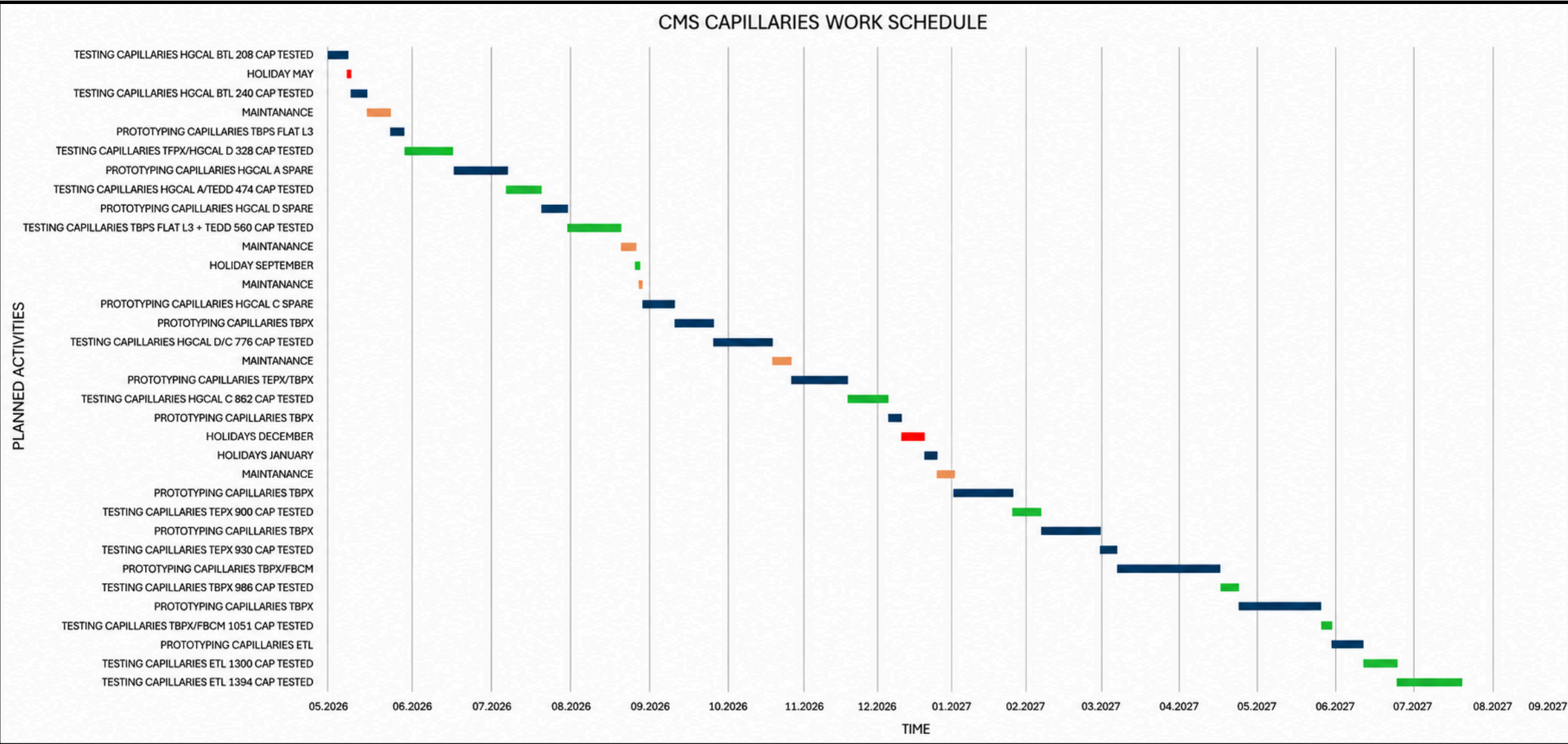
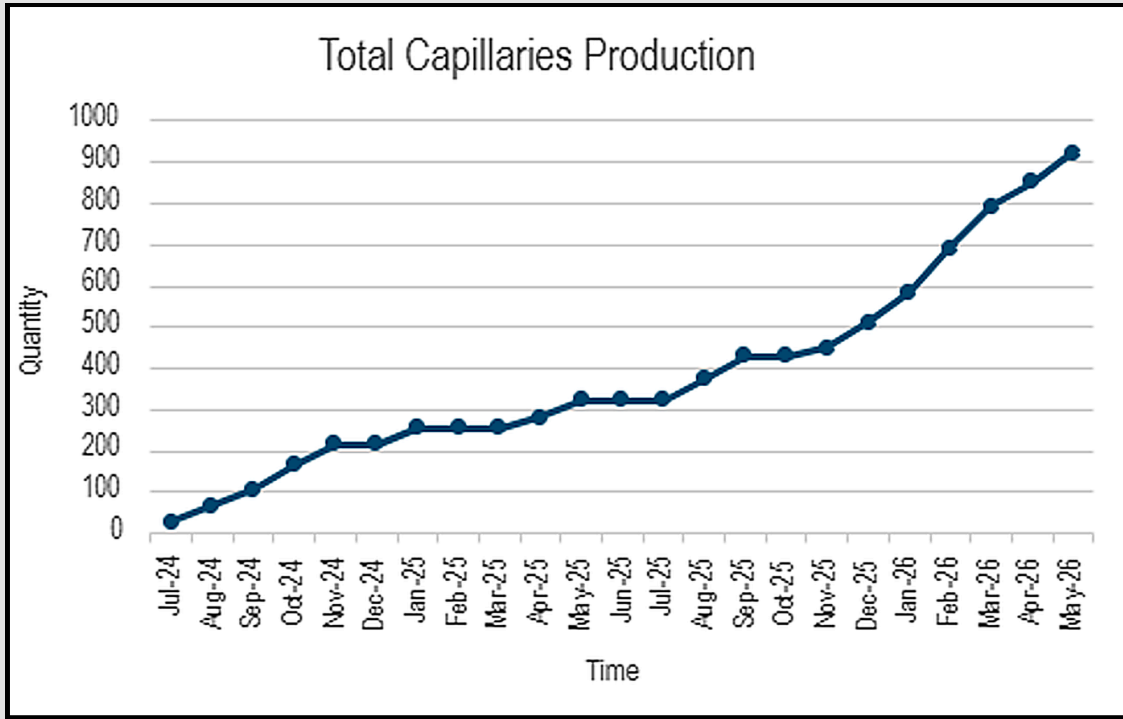
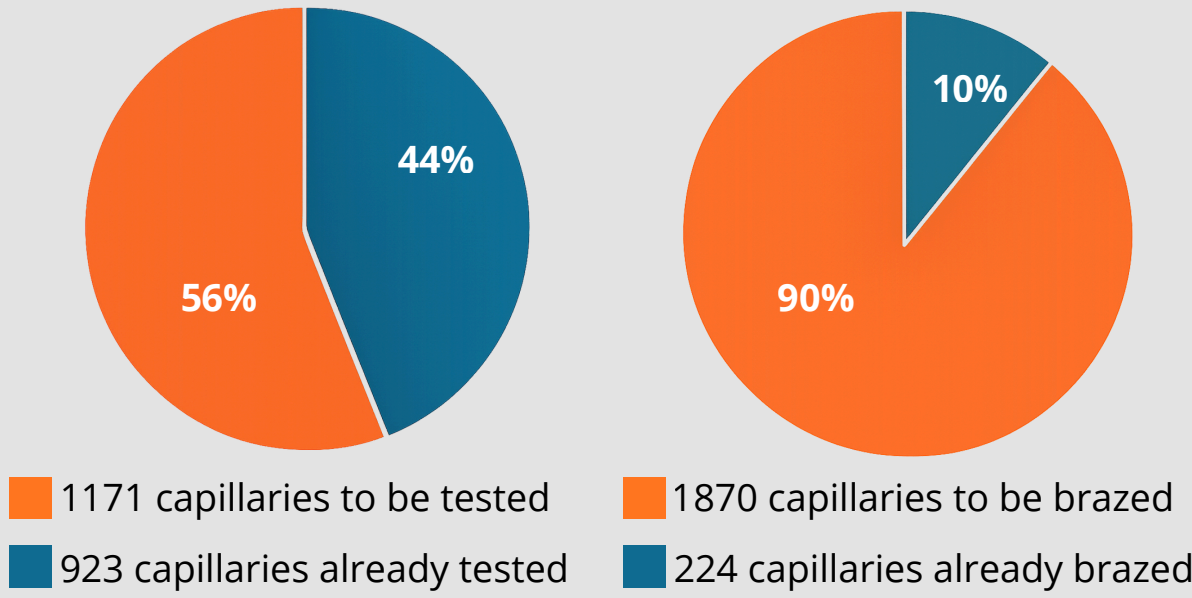
## RESULTS & OUTCOME

DP sensor error: +/-0,05%  
Flow meter sensor error: +/-0,20%

| CMS SUB-DETECTRS CAPILLARIES OVERVIEW |            |          |            |          |                   |                   |
|---------------------------------------|------------|----------|------------|----------|-------------------|-------------------|
| NAME                                  | INPUT DATA |          | TEST DATA  |          | ADDITIONAL VALUES |                   |
|                                       | FLOW (g/s) | DP (bar) | FLOW (g/s) | DP (bar) | DP st.dev (bar)   | FLOW st.dev (g/s) |
| *BTL                                  | 2,65       | 9,45     | 2,60       | 9,46     | 0,02              | 0,02              |
| *HGCAL A                              | 3,50       | 9,50     | 3,46       | 9,50     | 0,02              | 0,04              |
| HGCAL B                               | 4,00       | 9,50     | 4,10       | 9,51     | 0,03              | 0,03              |
| *HGCAL C                              | 2,30       | 9,50     | 2,28       | 9,51     | 0,03              | 0,01              |
| TB2S                                  | 1,26       | 8,80     | 1,26       | 8,79     | 0,12              | 0,05              |
| TBPS TILTED L1                        | 1,61       | 9,10     | 1,62       | 9,11     | 0,02              | 0,02              |
| TBPS TILTED L2                        | 2,16       | 8,10     | 2,17       | 8,09     | 0,04              | 0,02              |
| TBPS TILTED L3                        | 2,83       | 8,40     | 2,85       | 8,40     | 0,02              | 0,04              |
| *TEDD 1                               | 1,88       | 8,67     | 1,88       | 8,67     | 0,02              | 0,03              |
| *TEDD 2                               | 1,59       | 8,83     | 1,54       | 8,83     | 0,01              | 0,01              |

\* ongoing process of production

TBPX, TFPX, TBPS FLAT L1, TBPS FLAT L2 - prototyping phase



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CMS capillaries EDMS folder  
<https://edms.cern.ch/project/CERN-0000270811>